

Introduction to Yahtzee and Game Objective

Overview of Yahtzee:

Yahtzee is a popular dice game where players aim to score points by rolling five dice and achieving specific combinations. In each round, players can roll the dice up to three times, selecting which dice to keep or re-roll to improve their score in categories like Full House, Small Straight, or Yahtzee. The objective is to maximize the total score over 13 rounds by achieving the best possible combinations.

Assignment Objective:

This assignment focuses on developing a two-player version of Yahtzee in C programming. In this version, the human player (Player 1) competes against computer (Player 2), designed to play optimally. The task involves implementing game logic, score calculation, and a decision-making strategy for the AI to maximize its score.

Strategy for the Computer Player

Optimal Strategy Overview:

The AI is programmed to follow an optimal strategy, analyzing the dice rolls each turn to score the maximum possible points. It evaluates potential scoring categories based on the current dice state, aiming to make decisions that maximize its overall score.

Decision-Making Process:

After each roll, the AI assesses which dice to keep and which to re-roll to achieve the highest score. This involves calculating the probability of achieving higher scores various categories and adjusting its decisions based on these assessments. For instance, if the AI rolls two pairs, it may decide to re-roll one die to try for a Full House or even a higher combination, depending on the round's progress.

Scoring Category Selection:

Once the AI completes its rolls, it evaluates the dice's final state and selects the category that will yield the most points. This selection process takes into account unscored categories and those with the highest potential points. The AI's logic thus ensures it maximizes its score each round based on the dice configuration.

Implementation Challenges and Solutions

- I used chat gpt to get rid off some problems in functions and some time library.

Challenge 1 – Managing Game Logic:

Maintaining the game flow with multiple rolls, re-roll choices, and scoring calculations posed a significant challenge. Implementing these features while ensuring accurate scorekeeping and smooth transitions between turns required careful planning.

Solution:

Various functions and control structures were implemented to manage these processes effectively. Modular functions were created for rolling dice, choosing dice to re-roll, and calculating scores, making the game logic more manageable and organized. This modularity helped in debugging and refining specific parts of the code without disrupting other game functions.

Challenge 2 – AI Strategy Development:

Creating an effective strategy for the AI player was another significant hurdle. Programming the AI to analyse dice combinations and make decisions that would maximize its score required an understanding of the game's strategic aspects.

Solution:

To address this, different states of the dice were analysed to create a set of conditional strategies for the AI. By simulating various scenarios and possible outcomes, the AI was programmed to make optimal decisions that improved its chance of winning, taking into account probabilities and potential scores in each category.

Challenge 3 - Ensuring User Interactivity:

Providing clear and intuitive prompts for the human player was essential to enhance user experience. This involved designing user prompts for each stage of the game, including re-roll choices, scoring, and feedback on the AI's moves.

Solution:

Detailed prompts were created to guide the player through each decision-making step, and feedback was displayed to keep the player informed of the game's progress. This approach made it easier for users to interact with the game, while also improving the flow of the gameplay experience.

Reflection on Features Not Implemented**Optional or Missed Features:**

Additional features, such as graphical interfaces, were considered to improve user experience and visual appeal. However, these were not implemented due to time constraints and the added complexity they would bring to the current version of the game.

Conclusion**Summary of Achievement:**

The assignment challenges were addressed effectively, resulting in a functional two-player version of Yahtzee. Through structured functions and optimized AI decision-making, a fully playable game was developed that met the assignment's requirements.

Skills Developed:

Completing this assignment enhanced skills in C programming, especially in areas such as managing arrays, control structures, and implementing game logic. Working on AI strategies also deepened understanding of decision-making processes in programming, valuable skills applicable to broader problem-solving in software development.

Reported By

Krishnarajan Viyashan

IT 24103620